Weekly Homework 3

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Problem 1. Interpolate the rational function $f(x) = (1 + 25x^2)^{-1}$ over $[-1, 1]$

(a) using polynomials with a few equispaced nodes;

(b) using polynomials with a few Chebyshev nodes;

(c) using Hermite spline with a few equispaced nodes;

(d) using complete cubic spline with a few equispaced nodes;

(e) using natural cubic spline with a few equispaced nodes.

Visualize the results and estimate the interpolation error.

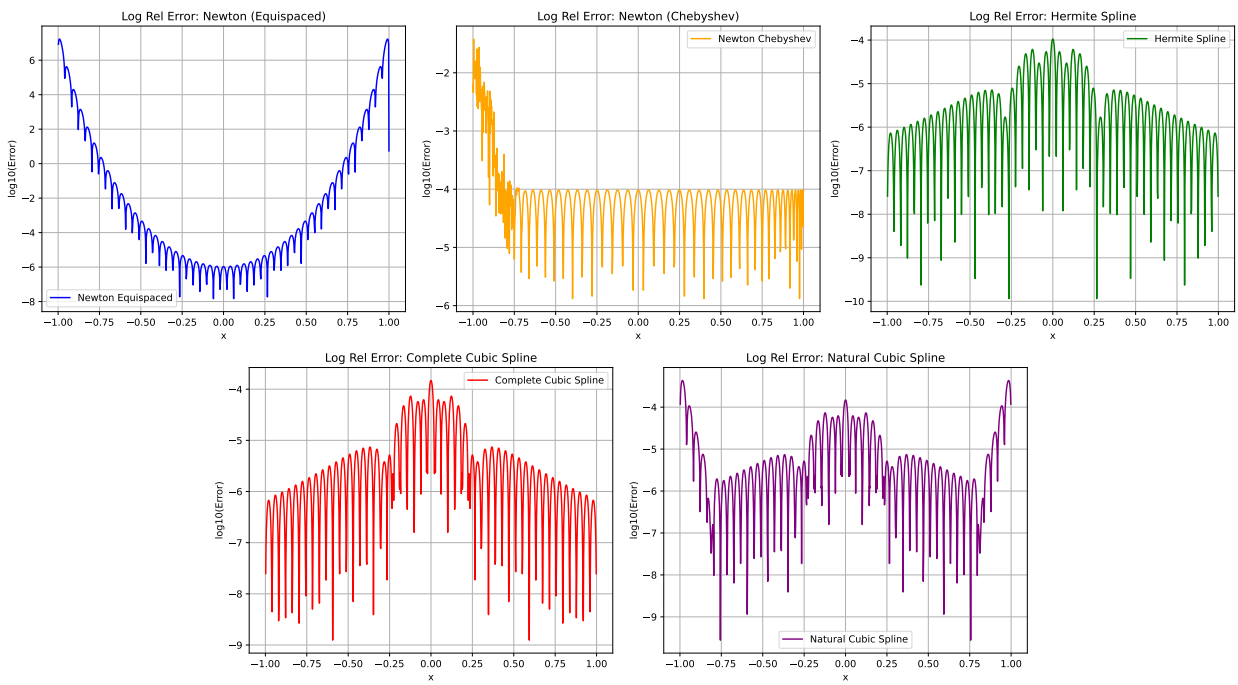

Figure 1: Interpolation error for different methods.

Problem 2. Let $f : \mathbb{R} \to \mathbb{R}$ be a sufficiently smooth function. Show that

$$\lim_{(x_0,x_1,\ldots,x_k)\to(x_*,x_*,\ldots,x_*)} f[x_0, x_1, \ldots, x_k] = \frac{f^{(k)}(x_*)}{k!}$$

for any $x_* \in \mathbb{R}$.

Proof. Let's first prove that

$$f[x_0, x_1, \ldots, x_k] = \frac{f^{(k)}(\xi)}{k!}, \quad \exists \xi \in [\min\{x_0, x_1, \ldots, x_k\}, \max\{x_0, x_1, \ldots, x_k\}]$$

for any distinct $x_0, x_1, \ldots, x_k$. Construct the function

$$g(t) = f(t) - s_k(t),$$

where $s_k(t)$ is the interpolation polynomial of f at the nodes $x_0, x_1, \ldots, x_k$. Then we have

$$g(x_i) = 0, \quad i = 0, 1, \ldots, k.$$

By Rolle's theorem, there exist k roots of g' in the interval $[\min\{x_0, x_1, \ldots, x_k\}, \max\{x_0, x_1, \ldots, x_k\}]$. Applying Rolle's theorem repeatedly, we can find a point ξ such that

$$g^{(k)}(\xi) = 0.$$

Since s_k is a polynomial of degree at most k, we have

$$s^{(k)}(t) = n! f[x_0, x_1, \ldots, x_k].$$

Therefore,

$$f^{(k)}(\xi) = s^{(k)}(\xi) = k! f[x_0, x_1, \ldots, x_k].$$

Hence,

$$f[x_0, x_1, \ldots, x_k] = \frac{f^{(k)}(\xi)}{k!}.$$

When $x_0, x_1, \ldots, x_k$ all approach x_*, the point ξ also approaches x_*. Therefore,

$$\lim_{(x_0,x_1,\ldots,x_k)\to(x_*,x_*,\ldots,x_*)} f[x_0, x_1, \ldots, x_k] = \lim_{\xi\to x_*} \frac{f^{(k)}(\xi)}{k!} = \frac{f^{(k)}(x_*)}{k!}.$$

□

Problem 3. The temperature in the human body is not a constant, but rather follows a daily rhythm driven by an internal biological clock. The following table lists 20 averaged values of temperature measurements taken from 70 English sailors in an experiment done in 1971.

Interpolate the data with a periodic cubic spline and plot your solution for a two-day-period.

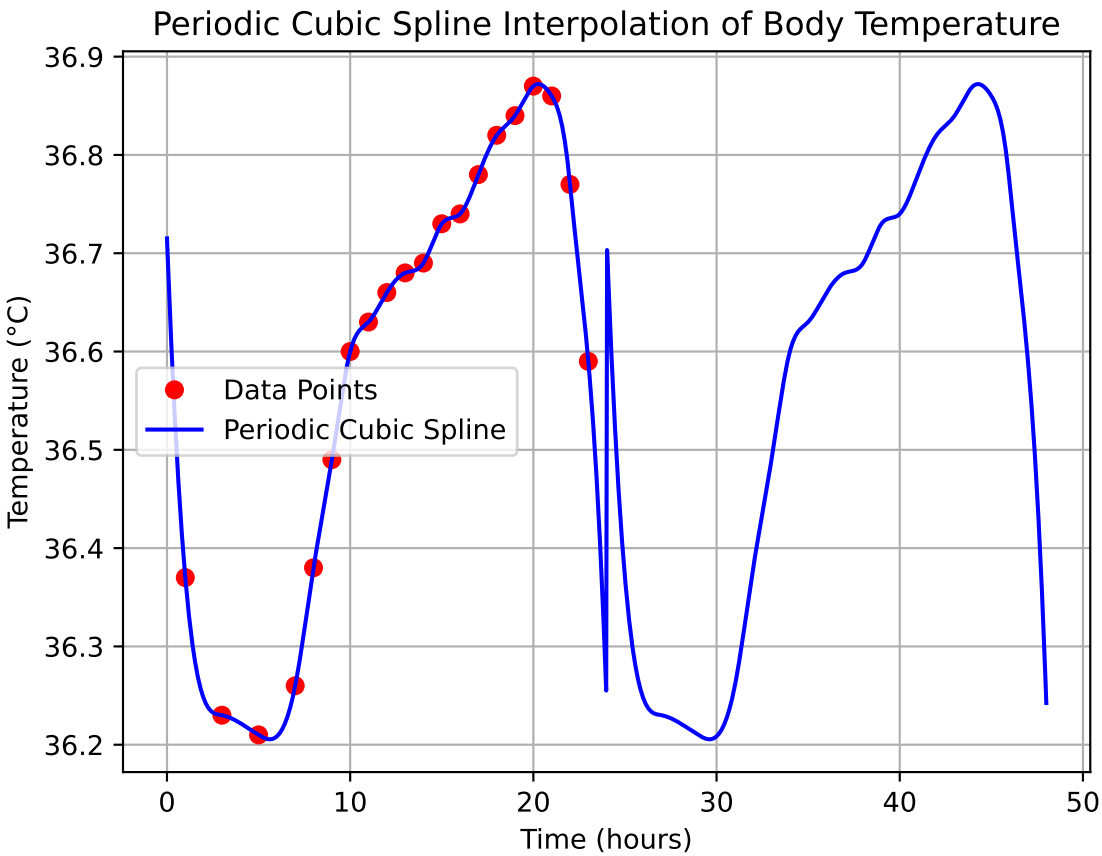

Figure 2: Periodic cubic spline interpolation of the temperature data.

Problem 4. When performing interpolation with a complete cubic spline, the choice of derivatives on the boundary is important. Suppose that Bob wants to interpolate the sine function $f(x) = \sin x$ at nine equispaced nodes over $[0, 2\pi]$, with $f'(0) = f'(2\pi) = 1$. nfortunately, he made a typo on $f'(0)$ in his program and observed some strange results. Try to reproduce Bob's result with a few different values of $f'(0)$. For instance, $f'(0) = 0$, $f'(0) = -1$, etc.

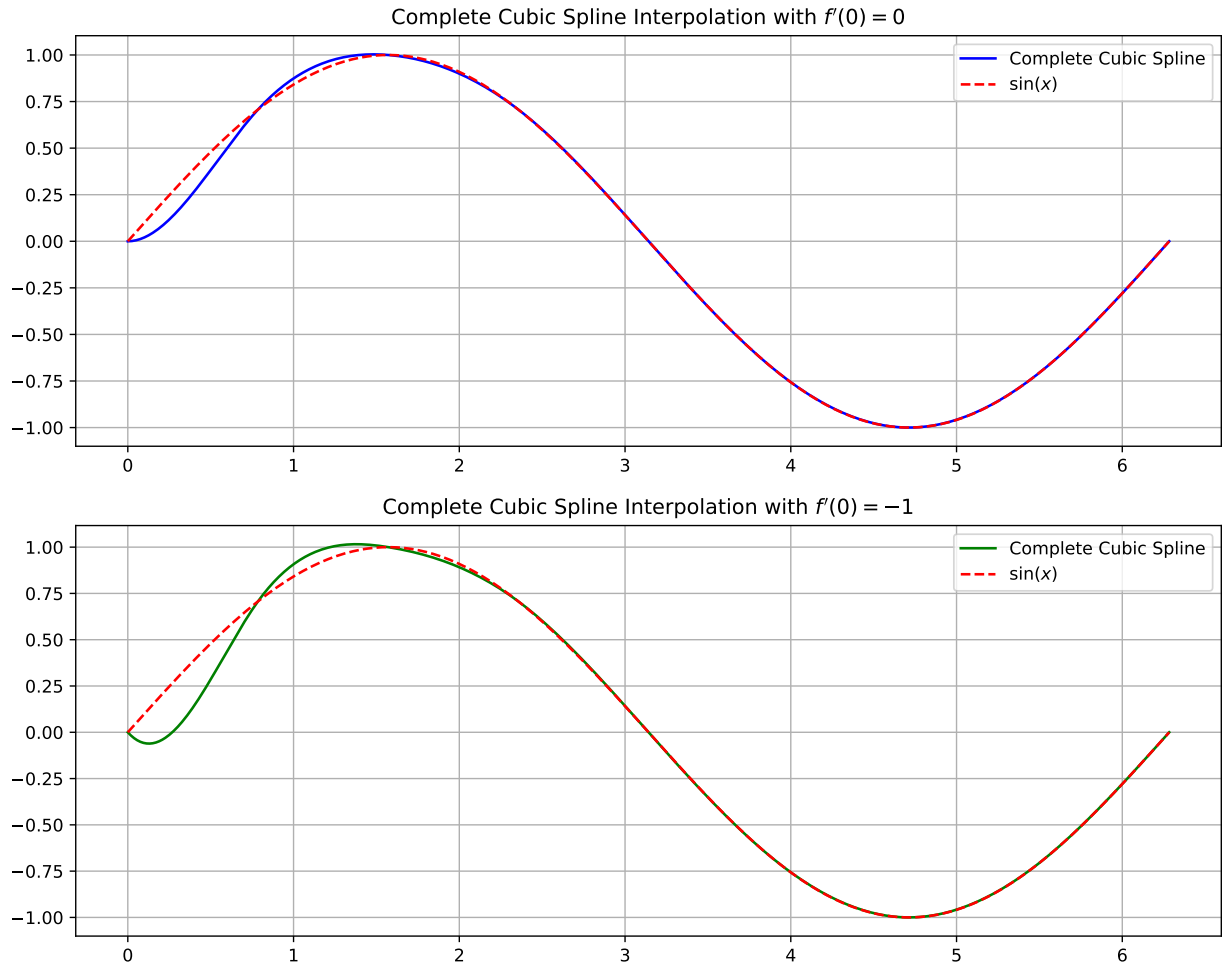

Figure 3: Complete cubic spline interpolation of the sine function with different boundary derivatives.

Problem 5. Similar to complete, natural, and periodic cubic splines, when the "not-a-knot" condition is used in cubic spline interpolation, the computational kernel is also to solve a sparse linear system. Try to derive the corresponding linear system.

Proof. The "not-a-knot" condition requires

$$s_0'''(x_1) = s_1'''(x_1), \quad s_{n-2}'''(x_{n-1}) = s_{n-1}'''(x_{n-1}),$$

which is equivalent to

$$\begin{aligned} -4\beta_0\eta_0 + 6\beta_0^2(k_0 + k_1) &= -4\beta_1\eta_1 + 6\beta_1^2(k_1 + k_2), \\ -4\beta_{n-2}\eta_{n-2} + 6\beta_{n-2}^2(k_{n-2} + k_{n-1}) &= -4\beta_{n-1}\eta_{n-1} + 6\beta_{n-1}^2(k_{n-1} + k_n). \end{aligned}$$

This is equivalent to

$$3\beta_0 k_0 + 3\left(\beta_0 - \frac{\beta_1^2}{\beta_0}\right) k_1 - 3\frac{\beta_1^2}{\beta_0} k_2 = 2\left(\eta_0 - \frac{\beta_1}{\beta_0}\eta_1\right),$$

$$3\beta_{n-2} k_{n-2} + 3\left(\beta_{n-2} - \frac{\beta_{n-1}^2}{\beta_{n-2}}\right) k_{n-1} - 3\frac{\beta_{n-1}^2}{\beta_{n-2}} k_n = 2\left(\eta_{n-2} - \frac{\beta_{n-1}}{\beta_{n-2}}\eta_{n-1}\right).$$

Along with the standard cubic spline equations, we can derive a linear system:

$$\begin{bmatrix} 3\beta_0 & 3\left(\beta_0 - \frac{\beta_1^2}{\beta_0}\right) & -3\frac{\beta_1^2}{\beta_0} & & & \\ \beta_0 & \alpha_1 & \beta_1 & & & \\ 0 & \beta_1 & \alpha_2 & \beta_2 & & \\ & & \ddots & \ddots & \ddots & \\ & & & \beta_{n-3} & \alpha_{n-2} & \beta_{n-2} \\ & & & 0 & \beta_{n-2} & 3\left(\beta_{n-2} - \frac{\beta_{n-1}^2}{\beta_{n-2}}\right) \end{bmatrix} \begin{bmatrix} k_0 \\ k_1 \\ k_2 \\ \vdots \\ k_{n-2} \\ k_{n-1} \\ k_n \end{bmatrix} = \begin{bmatrix} 2\left(\eta_0 - \frac{\beta_1}{\beta_0}\eta_1\right) \\ \eta_0 + \eta_1 \\ \eta_1 + \eta_2 \\ \vdots \\ \eta_{n-3} + \eta_{n-2} \\ 2\left(\eta_{n-2} - \frac{\beta_{n-1}}{\beta_{n-2}}\eta_{n-1}\right) \end{bmatrix}$$

$\square$